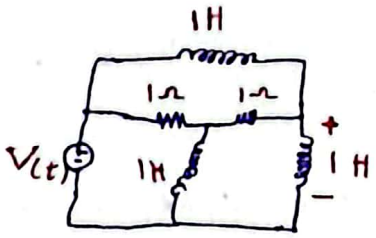
◀ سیستمهای زیر را مدل کنید:



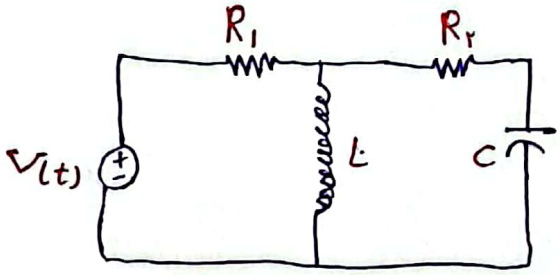
$$u(t)=i_1R_1+L_1\frac{di_1}{dt}+L_2\frac{d}{dt}(i_1-i_2)$$

$$0=i_2R_2+L_3\frac{di_2}{dt}+L_2\frac{d}{dt}(i_2-i_1)+V_L(t) \quad -1$$

$$u_L(t)=L_3\frac{di_2}{dt}$$

$$\begin{cases} \dot{i}_1=-i_1+u(t)-\dot{i}_2 \\ [\text{Pt}]\ \dot{i}_2=-i_2+i_1-u_L(t) \\ [\text{Pt}]\ u_L(t)=\dot{i}_2 \end{cases}$$

$$\begin{bmatrix}\dot{i}_1\\ \dot{i}_2\end{bmatrix}=\begin{bmatrix}-1 & 1\\ 1 & -2\end{bmatrix}\begin{bmatrix}i_1\\ i_2\end{bmatrix}+\begin{bmatrix}1\\ 0\end{bmatrix}u(t)\ ;\ u_L(t)=\begin{bmatrix}1 & -2\end{bmatrix}\begin{bmatrix}i_1\\ i_2\end{bmatrix}$$

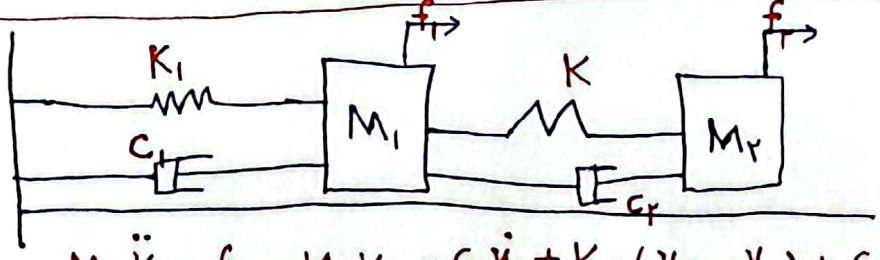


$$u(t)=R_1i_1+L\frac{d}{dt}(i_1-i_2) \quad -2$$

$$0=R_2i_2+L\frac{d}{dt}(i_2-i_1)+u_C(t)$$

$$i_2=C\frac{dv_C}{dt}$$

$$\begin{cases} L\dot{i}_1=u(t)-R_1i_1+L\dot{i}_2 \\ [\text{Pt}]\ L\dot{i}_2=-R_2i_2-u_C+L\dot{i}_1 \\ [\text{Pt}]\ C\dot{u}_C=i_2 \end{cases}$$



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$$M_1\ddot{x}_1=f_1-K_1x_1-c_1\dot{x}_1+K_2(x_2-x_1)+c_2(\dot{x}_2-\dot{x}_1)$$

$$M_2\ddot{x}_2=f_2-K(x_2-x_1)-c_2(\dot{x}_2-\dot{x}_1)$$

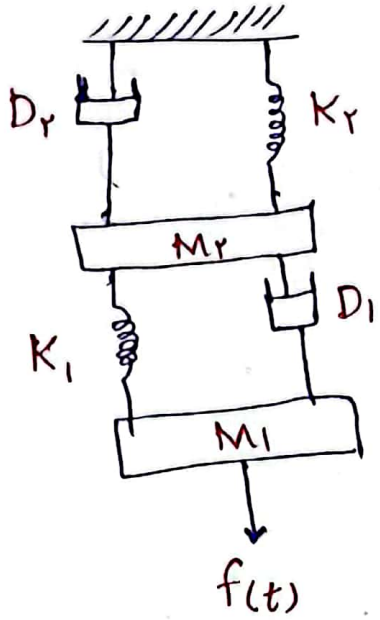
$$M_1\ddot{x}_1+(c_1+c_2)\dot{x}_1-c_2\dot{x}_2+(K_1+K_2)x_1-K_2x_2=f_1$$

$$\begin{cases} M_1\ddot{x}_1+(c_1+c_2)\dot{x}_1-c_2\dot{x}_2+(K_1+K_2)x_1-K_2x_2=f_1 \\ [\text{Pt}]\ M_2\ddot{x}_2-c_2\dot{x}_1+c_2\dot{x}_2-K_2x_1+K_2x_2=f_2 \end{cases}$$

(1)

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$$M_2\ddot{x}_2 = -K_2x_2 - D_2\dot{x}_2 + K_1(x_1 - x_2) + D_1(\dot{x}_1 - \dot{x}_2)$$

$$M_1\ddot{x}_1 = f(t) - K_1(x_1 - x_2) - D_1(\dot{x}_1 - \dot{x}_2)$$



$$M_1\ddot{x}_1 + D_1\dot{x}_1 - D_1\dot{x}_2 + K_1x_1 - K_1x_2 = f(t)$$

$$[Fpt]\,M_2\ddot{x}_2 - D_1\dot{x}_1 + (x_1 + D_2)\dot{x}_2 + K_1x_1 + (K_1 + K_2)x_2 = 0$$

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_1}{M_1} & -\frac{D_1}{M_1} & \frac{K_1}{M_1} & \frac{D_1}{M_1} \\ 0 & 0 & 0 & 1 \\ \frac{K_1}{M_2} & \frac{D_1}{M_2} & \frac{-K_1+K_2}{M_2} & \frac{-D_1+D_2}{M_2} \end{bmatrix} x + \begin{bmatrix} 0 \\ \frac{1}{M_1} \\ 0 \\ 0 \end{bmatrix} f(t)$$

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